

Advanced (Habanero)

- Q1.** What is the purpose of a scatter plot?
- (A) To show the relationship between two variables
 - (B) To display a single data point
 - (C) To create a line graph
 - (D) To make predictions
 - (E) To identify outliers
- Q2.** Which type of correlation is shown by the equation $y = 2x + 1$?
- (A) Negative
 - (B) Positive
 - (C) None
 - (D) Linear
 - (E) Quadratic
- Q3.** What is the line of best fit used for?
- (A) To find the median of a dataset
 - (B) To create a scatter plot
 - (C) To make predictions based on data trends
 - (D) To identify the mode of a dataset
 - (E) To calculate the range of a dataset
- Q4.** If a scatter plot shows a negative correlation, what happens to y as x increases?
- (A) y increases
 - (B) y decreases
 - (C) y remains constant
 - (D) y becomes zero
 - (E) y becomes undefined
- Q5.** Which of the following is an example of a positive correlation?
- (A) The more hours studied, the lower the grade
 - (B) The more hours studied, the higher the grade
 - (C) The more hours slept, the more hours studied
 - (D) The more hours watched TV, the less hours slept
 - (E) The more hours exercised, the less hours slept
- Q6.** What is the purpose of identifying outliers in a scatter plot?
- (A) To remove them from the dataset
 - (B) To add more data points
 - (C) To change the scale of the axes
 - (D) To identify patterns or trends
 - (E) To calculate the mean of the dataset
- Q7.** If the equation of the line of best fit is $y = 3x - 2$, what is the y -intercept?
- (A) -2
 - (B) 3
 - (C) 2
 - (D) -3
 - (E) 1
- Q8.** What is the slope of the line of best fit in the equation $y = 2x + 1$?
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
 - (E) 5

Open Ended Questions (FRQ)

- Q9.** Given the data points $(1, 2)$, $(2, 3)$, $(3, 5)$, and $(4, 7)$, create a scatter plot and draw the line of best fit. Then, use the line of best fit to predict the value of y when $x = 5$. Show all work and explain your reasoning.
- Q10.** Explain the differences between a positive, negative, and no correlation. Provide examples of each and explain how the line of best fit is used in each case. Be sure to include equations and graphs to support your answer.

Chef's Tips

- Emphasize the importance of labeling axes and including a title in the scatter plot
- Remind students to show all work and explain their reasoning in the free response questions
- Encourage students to use technology, such as graphing calculators or computer software, to create scatter plots and lines of best fit